CHAPTER VII

The Propositional Calculus

Words and Symbols

THE PRECEDING DIALOGUE is reminiscent of the *Two-Part Invention* by Lewis Carroll. In both, the Tortoise refuses to use normal, ordinary in the normal, ordinary way-or at least he refuses to do so when it is his advantage to do so. A way to think about the Carroll paradox was given last Chapter. In this Chapter we are going to make symbols dc Achilles couldn't make the Tortoise do with his words. That is, we are to make a formal system one of whose symbols will do just what A wished the word `and' would do, when spoken by the Tortoise, and ail of whose symbols will behave the way the words *'if... then* . . .' ought to behave. There are only two other words which we will attempt to deal with `or' and `not'. Reasoning which depends only on correct usage of these words is termed *propositional reasoning*.

Alphabet and First Rule of the Propositional Calculus

I will present this new formal system, called the Propositional Calculus, like a puzzle, not explaining everything at once, but letting you things out to some extent. We begin with the list of symbols:

	<	>	
P	**Q**	**R**	′
∧	∨	⊃	~

The first rule of this system that I will reveal is the following:

RULE OF JOINING: If x and y are theorems of the system, then so is the string < x∧y >.

This rule takes two theorems and combines them into one. It s remind you of the Dialogue.

Well-Formed Strings

There will be several other rules of inference, and they will all be pres shortly-but first, it is important to define a subset of all strings, namely the

well formed strings. They will be defined in a recursive way. We begin with the
ATOMS: **P, Q,** and **R** are called atoms.. New atoms are formed by appending primes onto the right of old atoms-thus, **R', Q'', P'''**, etc. This gives an endless supply of atoms. All atoms are well-formed.

Then we have four recursive

FORMATION RULES: If x and y are well-formed, then the following four strings are also well-formed:

(1) ~x
(2) < x∧y>
(3) < x∨y>
(4) < x⊃y>

For example, all of the following are well-formed:

P	atom
~P	by (1)
~~P	by (1)
Q´	atom
~Q1	by (1)
<P∧~Q' >	by (2)
~<P∧~Q' >	by (1)
~~<P⊃~Q' >	by (4)
<~<P∧~Q' >∨~~<P⊃~Q' >>	by (3)

The last one may look quite formidable, but it is built up straightforwardly from two components-namely the two lines just above it. Each of them is in turn built up from previous lines ... and so on. Every well-formed string can in this way be traced back to its elementary constituents-that is, atoms. You simply run the formation rules backwards until you can no more. This process is guaranteed to terminate, since each formation rule (when run forwards) is a *lengthening* rule, so that running it backwards always drives you towards atoms.

This method of decomposing strings thus serves as a check on the well-formedness of any string. It is a *top-down decision procedure* for wellformedness. You can test your understanding of this decision procedure by checking which of the following strings are well-formed:

(1) <P>
(2) (2) <~P>
(3) <P∧Q∧R>
(4) <P∧Q>
(5) <<P∧Q>∧Q~∧P>>
(6) <P∧~P>
(7) <<P∨<Q⊃R>>∧<~P∨~R´>>
(8) <P∧Q>∧< Q∧P:

(Answer: Those whose numbers are Fibonacci numbers are not formed. The rest are well-formed.)

More Rules of Inference

Now we come to the rest of the rules by which theorems of this system constructed. A few rules of inference follow. In all of them, the symbols ´x´ and 'y' are always to be understood as restricted to *well formed* strings

RULE OF SEPARATION: If < x∧y> is a theorem, then both x and theorems.

Incidentally, you should have a pretty good guess by now as to concept the symbol `A' stands for. (Hint: it is the troublesome word the preceding Dialogue.) From the following rule, you should be a figure out what concept the *tilde* ('~') represents:

DOUBLE-TILDE RULE: The string '~~' can be deleted from any theorem. It can also be inserted into any theorem, provided that the rest string is itself well-formed.

The Fantasy Rule

Now a special feature of this system is that it has *no axioms*-only rule you think back to the previous formal systems we've seen, you may w(how there can be any theorems, then. How does everything get started? The answer is that there is one rule which manufactures theorems from out of thin air-it doesn't need an "old theorem" as input. (The rest of the do require input.) This special rule is called the *fantasy rule*. The reason I call it that is quite simple.

To use the fantasy rule, the first thing you do is to write down an well-formed string x you like, and then "fantasize" by asking, "*What if* string x were an axiom, or a theorem?" And then, you let the system give an answer. That is, you go ahead and make a derivation with x ; opening line; let us suppose y is the last line. (Of course the derivation must strictly follow the rules of the system.) Everything from x to y (inclusive) is the *fantasy*; x is the *premise* of the fantasy, and y is its *outcome*. The next step is to jump out of the fantasy, having learned from it that out.

If x were a theorem, y would be a theorem.

Still, you might wonder, where is the *real* theorem? The real theorem is the string

<x⊃y>

Notice the resemblance of this string to the sentence printed above

To signal the entry into, and emergence from, a fantasy, one uses the

square brackets `[' and ']', respectively. Thus, whenever you see a left square bracket, you know you are "pushing" into a fantasy, and the *next* line will contain the fantasy's *premise*. Whenever you see a right square bracket, you know you are "popping" back out, and the *preceding* line was the outcome. It is helpful (though not necessary) to indent those lines of a derivation which take place in fantasies.

Here is an illustration of the fantasy rule, in which the string P is taken as a premise. (It so happens that P is not a theorem, but that is of no import; we are merely inquiring, "What if it were?") We make the following fantasy:

[	push into fantasy
P	premise
~~~P	outcome (by double tilde rule)
]	pop out of fantasy

The fantasy shows that:

If **P** were a theorem, so would **~~P** be one.

We now "squeeze" this sentence of English (the metalanguage) into the formal notation (the object language): <P⊃~~P>. This, our first theorem of the Propositional Calculus, should reveal to you the intended interpretation of the symbol `⊃'.

Here is another derivation using the fantasy rule:

[	push
<P∧Q>	premise
P	separation
Q	separation
<Q∧P>	joining
]	pop
<<P∧Q>⊃<Q∧P>>	fantasy rule

It is important to understand that only the last line is a genuine theorem, here-everything else is in the fantasy.

Recursion and the Fantasy Rule

As you might guess from the recursion terminology "push" and "pop", the fantasy rule can be used recursively-thus, there can be fantasies within fantasies, thrice-nested fantasies, and so on. This means that there are all sorts of "levels of reality", just as in nested stories or movies. When you pop out of a movie-within-a-movie, you feel for a moment as if you had reached the real world, though you are still one level away from the top. Similarly, when you pop out of a fantasy-within-a-fantasy, you are in a "realer" world than you had been, but you are still one level away from the top.

Now a "No Smoking" sign inside a movie theater does not apply to the

characters in the movie-there is no carry-over from the real world in fantasy world, in movies. But in the Propositional Calculus, then carry-over from the real world into the fantasies; there is even carry from a fantasy to fantasies inside it. This is formalized by the following rule:

CARRY-OVER RULE: Inside a fantasy, any theorem from the "reality level higher can be brought in and used.

It is as if a "No Smoking" sign in a theater applied not only to a moviegoers, but also to all the actors in the movie, and, by repetition of the same idea, to anyone inside multiply nested movies! (Warning: There carry-over in the reverse direction: theorems inside fantasies cannot be exported to the exterior! If it weren't for this fact, you could write any as the first line of a fantasy, and then lift it out into the real world as a theorem.)

To show how carry-over works, and to show how the fantasy rule can be used recursively, we present the following derivation:

[	push
P	premise of outer fantasy
[	push again
Q	premise of inner fantasy
P	carry-over of P into inner fantasy
<P∧Q>	joining
]	pop out of inner fantasy, regain outer fantasy
<Q⊃<P∧Q>>	fantasy rule
]	pop out of outer fantasy, reach real world!
<P⊃<Q⊃<P∧Q>>>	fantasy rule

Note that I've indented the outer fantasy once, and the inner fantasy twice, to emphasize the nature of these nested "levels of reality". One to look at the fantasy rule is to say that an observation made *about* the system is inserted *into* the system. Namely, the theorem < x⊃y> which gets produced can be thought of as a representation inside the system of the statement about the system "If x is a theorem, then y is too". To be specific, the intended interpretation for **<P⊃Q>** is "if **P**, then **Q** equivalently, "**P** implies **Q**".

The Converse of the Fantasy Rule

Now Lewis Carroll's Dialogue was all about "if-then" statements. In particular, Achilles had a lot of trouble in persuading the Tortoise to accept the second clause of an "if-then" statement, even when the "if-then" state itself was accepted, as well as its first clause. The next rule allows y infer the second "clause" of a'⊃'-string, provided that the `⊃'-string it a theorem, and that its first "clause" is also a theorem.

RULE OF DETACHMENT: If x and < x⊃y> are both theorems, then y is a theorem.

Incidentally, this rule is often called "Modus Ponens", and the fantasy rule is often called the "Deduction Theorem".

The Intended Interpretation of the Symbols

We might as well let the cat out of the bag at this point, and reveal the "meanings" of the rest of the symbols of our new system. In case it is not yet apparent, the symbol `A' is meant to be acting isomorphically to the normal, everyday word `and'. The symbol '-' represents the word 'not'-it is a formal sort of negation. The angle brackets '<' and `>' are groupers-their function being very similar to that of parentheses in ordinary algebra. The main difference is that in algebra, you have the freedom to insert parentheses or to leave them out, according to taste and style, whereas in a formal system, such anarchic freedom is not tolerated. The symbol '∨' represents the word `or' ('vel' is a Latin word for `or'). The `or' that is meant is the so-called *inclusive* `or', which means that the interpretation of <x∨y> is "either x or y-or both".

The only symbols we have not interpreted are the atoms. An atom has no single interpretation-it may be interpreted by any sentence of English (it must continue to be interpreted by the same sentence if it occurs multiply within a string or derivation). Thus, for example, the well-formed string <**P**∧~**P**> could be interpreted by the compound sentence

This mind is Buddha, and this mind is not Buddha.

Now let us look at each of the theorems so far derived, and interpret them. The first one was <**P**⊃~~**P**>. If we keep the same interpretation for **P**, we have the following interpretation:

If this mind is Buddha,
then it is not the case that this mind is not Buddha.

Note how I rendered the double negation. It is awkward to repeat a negation in any natural language, so one gets around it by using two different ways of expressing negation. The second theorem we derived was **<<P∧Q>⊃<Q∧P>>.** If we let **Q** be interpreted by the sentence "This flax weighs three pounds", then our theorem reads as follows:

If this mind is Buddha and this flax weighs three pounds,
then this flax weighs three pounds and this mind is Buddha.

The third theorem was <**P**⊃<**Q**⊃<**P**∧**Q**>>>. This one goes into the following nested "if-then" sentence:

If this mind is Buddha,
then, if this flax weighs three pounds,
then this mind is Buddha and this flax weighs three pounds.

You probably have noticed that each theorem, when interpreted, something absolutely trivial and self-evident. (Sometimes they are so s evident that they sound vacuous and-paradoxically enough-confusing or even wrong!) This may not be very impressive, but just remember there are plenty of falsities out there which could have been produced they weren't. This system-the Propositional Calculus-steps neatly ft truth to truth, carefully avoiding all falsities, just as a person who is concerned with staying dry will step carefully from one stepping-stone creek to the next, following the layout of stepping-stones no matter I twisted and tricky it might be. What is impressive is that-in the Propositional Calculus-the whole thing is done purely *typographically*. There is nobody down "in there", thinking about the *meaning* of the strings. It i! done mechanically, thoughtlessly, rigidly, even stupidly.

Rounding Out the List of Rules

We have not yet stated all the rules of the Propositional Calculus. The complete set of rules is listed below, including the three new ones.

JOINING RULE: If x and y are theorems, then < x∧y> is a theorem.

SEPARATION RULE: If < x∧y> is a theorem, then both x and y are theorems.

DOUBLE-TILDE RULE: The string '~~' can be deleted from any theorem can also be inserted into any theorem, provided that the result string is itself well-formed.

FANTASY RULE: If y can be derived when x is assumed to be a theorem then < x⊃y> is a theorem.

CARRY-OVER RULE: Inside a fantasy, any theorem from the "reality" c level higher can be brought in and used.

RULE OF DETACHMENT: If x and < x⊃y> are both theorems, then y is a theorem.

CONTRAPOSITIVE RULE: <x⊃y> and <~y⊃~x> are interchangeable

DE MORGAN'S RULE: <~x∧~y> and ~< x∨y> are interchangeable.

SWITCHEROO RULE: <x∨y> and <~x⊃y> are interchangeable.

(The Switcheroo rule is named after Q. q. Switcheroo, an Albanian railroad engineer who worked in logic on the siding.) By "interchangeable" in foregoing rules, the following is meant: If an expression of one form occurs as either a theorem or part of a theorem, the other form may be

substituted, and the resulting string will also be a theorem. It must be kept in mind that the symbols ‘x’ and ‘y’ always stand for well-formed strings of the system.

Justifying the Rules

Before we see these rules used inside derivations, let us look at some very short justifications for them. You can probably justify them to yourself better than my examples – which is why I only give a couple.

The contrapositive rule expresses explicitly a way of turning around conditional statements which we carry out unconsciously. For instance, the “Zentence”

If you are studying it, then you are far from the Way

Means the same thing as

If you are close to the Way, then you are not studying it.

De Morgan’s rule can be illustrated by our familiar sentence “The flag is not moving and the wind is not moving”. If **P** symbolizes “the flag is not moving”, and **Q** symbolizes “the wind is moving”, then the compound sentence is symbolized by <**~P∧~Q**>, which, according to Morgan’s law, is interchangeable with **~<P∨Q>.** whose interpretation would be “It is not true that either the flag or the wind is moving”. And no one could deny that it is a Zensible conclusion to draw.

For the Switrcheroo rule, consider the sentence “Either a cloud is hanging over the mountain, or the moonlight is penetrating the waves of the lake,” which might be spoken, I suppose, by a wistful Zen master remembering a familiar lake which he can visualize mentally but cannot see. Now hang on to your seat, for the Swircheroo rule tells us that this is interchangeable with the thought “If a cloud is not hanging over the mountain, then the moonlight is penetrating the waves of the lake.” This may not be enlightenment, but it is the best the Propositional Calculus has to offer.

Playing around with the system

Now, let us apply these rules to a previous theorem, ands see what we get: For instance, take the theorem **<P⊃~~P>**:

<P⊃~~P>:	old theorem
<~~~P⊃~P>:	contrapositive
<~P⊃~P>	double-tilde
<P∨~P>	switcheroo

This new theorem, when interpreted, says:

Either this mind is Buddha, or this mind is not Buddha

Once again, the interpreted theorem, though perhaps less than mind boggling, is at least true.

Semi-Interpretations

It is natural, when one reads theorems of the Propositional Calculus out loud, to interpret everything but the atoms. I call this semi-interpreting. For example, the semi-interpretation of **<P∨~P>**:: would be

P or not **P**.

Despite the fact that P is not a sentence, the above semisentence still sounds true, because you can very easily imagine sticking any sentence in for P – and the form of the semi-interpreted theorem assures you that however you make your choice, the resulting sentence will be true. And that is the key idea of the Propositional Calculus: it produces theorems which, when semi-interpreted, are seen to be "universally true semisaentences", by which is meant that no matter how you complete the interpretation, the final result will be a true statement.

Ganto's Ax

Now we can do a more advanced exercise, based on a Zen koan called "Ganto's Ax". Here is how it began.

> One day Tokusan told his student Ganto, "I have two monks who have been here for many years. Go and examine them." Ganto picked up an ax and went to the hut where the two monks were meditating. He raised the ax, saying "If you say a word, I will cut off your heads; and if you do not say a word, I will also cut off your heads."[1]

If you say a word I will cut off this koan, and if you do not say a word, I will also cut off this koan – because I want you to translate some of it into our notation. Let us symbolize "you say a word" by **P** and "I will cut off your heads" by **Q**. Then Ganto's ax threat is symbolized by the string **<<P⊃Q>∧<~`P⊃Q>>**. What if this ax threat were an axiom? Here is a fantasy to answer that question.

(1)	[	push
(2)	**<<P⊃Q>∧<~`P⊃Q>>**.	Ganto's axiom
(3)	**<P⊃Q>**	separation
(4)	**<~Q⊃~P>**.	contrapositive
(5)	<~**P⊃Q>**	separation
(6)	**<~Q⊃~~P>**.	contrapositive
(7)	]	push again
(8)	~**Q**	premise

(9)	**<~Q⊃~P>**.	carry-over of line 4
(10)	**~P**	detachment
(11)	**<~Q⊃~~P>**.	carry-over of line 6
(12)	**~~P**	detachment (lines 8 and 11)
(13)	**<~P∧~~P>**	joining
(14)	**<~P∨~~P>**	De Morgan
(15)	]	pop once
(16)	**<~Q⊃~<P∨~P>>**.	fantasy rule
(17)	**<~P∨~P>⊃Q>**.	contrapositive
(18)	[	push
(19)	. **~P**	premise (also outcome)
(20)	]	pop
(21)	**<~P⊃~P>**.	fantasy rule
(22)	**<P∨~P>**.	switcheroo
(23)	**Q**	detachment (lines 22 and 17)
(24)	]	pop out

The power of the Propositional Calculus is shown in this example. Why, in but two dozen steps, we have deduced Q: that the heads will be cut off! (Ominously, the rule last invoked was "detachment" ...) It might seem superfluous to continue the koan now, since we know what must ensue ... However, I shall drop my resolve to cut the koan off; it is a true Zen koan, after all. The rest of the incident is here related:

> Both monks continued their meditation as if he had not spoken. Ganto dropped the ax and said, "You are true Zen students." He returned to Tokusan and related the incident. "I see your side well," Tokusan agreed, "but tell me, how is their side?" "Tõzan may admit them," replied Ganto, "but they should not be admitted under Tokusan."2

Do you see my side well? How is the Zen side?

Is There a Decision Procedure for Theorems?

The Propositional Calculus gives us a set of rules for producing statements which would be true in all conceivable worlds. That is why all of its theorems sound so simple-minded; it seems that they have absolutely no content! Looked at this way, the Propositional Calculus might seem to be a waste of time, since what it tells us is absolutely trivial. On the other hand, it does it by specifying the *form* of statements that are universally true, and this throws a new kind of light onto the core truths of the universe: they are not only fundamental, but also *regular*: they can be produced by one set of typographical rules. To put it another way, they are all "cut from the same cloth". You might consider whether the same could be said about Zen koans: could they all be produced by one set of typographical rules?

It is quite relevant here to bring up the question of a decision procedure. That is, does there exist any mechanical method to tell nontheorems from theorems? If so, that would tell us that the set of theorems of the

Propositional Calculus is not only r.e., but also recursive. It turns out that there is an interesting decision procedure-the method of truth u would take us a bit afield to present it here; you can find it in almost any standard book on logic. And what about Zen koans? Could there conceivably be a mechanical decision procedure which distinguishes genuine Zen koans from other things?

Do We Know the System Is Consistent?

Up till now, we have only *presumed* that all theorems, when interpreted as indicated, are true statements. But do we *know* that that is the case' we prove it to be? This is just another way of asking whether the intended interpretations ('and' for `∧', etc.) merit being called the "passive meanings" of the symbols. One can look at this issue from two very different points of view, which might be called the "prudent" and "imprudent" points I will now present those two sides as I see them, personifying their as "Prudence" and "Imprudence".

Prudence: We will only KNOW that all theorems come out true un intended interpretation if we manage to PROVE it. That is the c: thoughtful way to proceed.

Imprudence: On the contrary. It is OBVIOUS that all theorems will come out true. If you doubt me, look again at the rules of the system. You will find that each rule makes a symbol act exactly as the word it represents ought to be used. For instance, the joining rule makes the symbol '∧' act as `and' ought to act; the rule of detachment makes `⊃' act as it ought to, if it is to stand for 'implies', or 'if-then'; and so on. Unless you are like the Tortoise, you will recognize in each rule a codification of a pattern you use in your own thought patterns. So if you trust your own thought patterns, then you HAVE to believe that all theorems come out true! That's the way I see it. I don't need any further proof. If you think that some theorem comes out false, then presumably you think that some rule must be wrong. Show me which one.

Prudence: I'm not sure that there is any faulty rule, so I can't point one out to you. Still, I can imagine the following kind of scenario. You, following the rules, come up with a theorem -- say x. Meanwhile I, also following the rules, come up with another theorem-it happens to be ~x. Can't you force yourself to conceive of that?

Imprudence: All right; let's suppose it happened. Why would it bother you? Or let me put it another way. Suppose that in playing with the **MIU**-system, I came up with a theorem x, and you came up with xU Can you force yourself to conceive of that?

Prudence: Of course-in fact both **MI** and **MIU** are theorems.

Imprudence: Doesn't that bother you?

Prudence: Of course not. Your example is ridiculous, because **MI** and **MIU** are not CONTRADICTORY, whereas two strings x and ~x in the Propositional Calculus ARE contradictory.

Imprudence: Well, yes -- provided you wish to interpret `~' as `not'. But what would lead you to think that '~' should be interpreted as `not'?

Prudence: The rules themselves. When you look at them, you realize that the only conceivable interpretation for '~' is 'not'-and likewise, the only conceivable interpretation for `∧' is `and', etc.

Imprudence: In other words, you are convinced that the rules capture the meanings of those words?

Prudence: Precisely.

Imprudence: And yet you are still willing to entertain the thought that both x and ~x could be theorems? Why not also entertain the notion that hedgehogs are frogs, or that 1 equals 2, or that the moon is made of green cheese? I for one am not prepared even to consider whether such basic ingredients of my thought processes are wrong -- because if I entertained that notion, then I would also have to consider whether my modes of analyzing the entire question are also wrong, and I would wind up in a total tangle.

Prudence: Your arguments are forceful ... Yet I would still like to see a PROOF that all theorems come out true, or that x and ~x can never both be theorems.

Imprudence: You want a proof. I guess that means that you want to be more convinced that the Propositional Calculus is consistent than you are convinced of your own sanity. Any proof I could think of would involve mental operations of a greater complexity than anything in the Propositional Calculus itself. So what would it prove? Your desire for a proof of consistency of the Propositional Calculus makes me think of someone who is learning English and insists on being given a dictionary which definers all the simple words in terms of complicated ones...

The Carroll Dialogue Again

This little debate shows the difficulty of trying to use logic and reasoning to defend themselves. At some point, you reach rock bottom, and there is no defense except loudly shouting, "I know I'm right!" Once again, we are up against the issue which Lewis Carroll so sharply set forth in his Dialogue: you can't go on defending your patterns of reasoning forever. There comes a point where faith takes over.

A system of reasoning can be compared to an egg. An egg has a shell which protects its insides. If you want to ship an egg somewhere, though, you don't rely on the shell. You pack the egg in some sort of container, chosen according to how rough you expect the egg's voyage to be. To be extra careful, you may put the egg inside several nested boxes. However, no matter how many layers of boxes you pack your egg in, you can imagine some cataclysm which could break the egg. But that doesn't mean that you'll never risk transporting your egg. Similarly, one can never give an ultimate, absolute proof that a proof in some system is correct. Of course,

one can give a proof of a proof, or a proof of a proof of a proof – but the validity of the outermost system always remains an unproven assumption, accepted on faith. One can always imagine that some unsuspected subtlety will invalidate every single level of proof down to the bottom, and tI "proven" result will be seen not to be correct after all. But that doesn't mean that mathematicians and logicians are constantly worrying that the whole edifice of mathematics might be wrong. On the other hand, unorthodox proofs are proposed, or extremely lengthy proofs, or proofs generated by computers, then people do stop to think a bit about what they really mean by that quasi-sacred word "proven".

An excellent exercise for you at this point would be to go back Carroll Dialogue, and code the various stages of the debate into our notation -- beginning with the original bone of contention:

Achilles: If you have **<<A∧B>⊃Z>,** and you also have **<A∧B>,** then surely you have Z.
Tortoise: Oh! You mean: **<<<<A∧B>⊃Z>∧<A∧B>>⊃Z>,** : don't you?

(Hint: Whatever Achilles considers a rule of inference, the Tortoise immediately flattens into a mere string of the system. If you use or letters **A, B**, and **Z**, you will get a recursive pattern of longer and strings.)

Shortcuts and Derived Rules

When carrying out derivations in the Propositional Calculus, one quickly invents various types of shortcut, which are not strictly part of the system For instance, if the string **<Q∨~Q>** were needed at some point, and **<P∨~P>** had been derived earlier, many people would proceed as if **<Q∨~Q>** had been derived, since they know that its derivation is an exact parallel to that of **<P∨~P>.** The derived theorem is treated as a "theorem schema" -- a mold for other theorems. This turns out to be a perfect valid procedure, in that it always leads you to new theorems, but it is not a rule of the Propositional Calculus as we presented it. It is, rather, a *derived rule*, It is part of the knowledge which we have *about* the system. That this rule keeps you within the space of theorems needs proof, of course- but such a proof is not like a derivation inside the system. It is a proof in the ordinary, intuitive sense -- a chain of reasoning carried out in the I-mode. The theory about the Propositional Calculus is a "metatheory", and results in it can be called "metatheorems" - Theorems about theorems. (Incidentally, note the peculiar capitalization in the phrase "Theorems about theorems". It is a consequence of our convention: metatheorems are Theorems (proven results) concerning theorems (derivable strings).)

In the Propositional Calculus, one could discover many metatheorems, or derived rules of inference. For instance, there is a De Morgan's Rule:

$<\sim x \vee \sim y>$ and $\sim<x \wedge y>$ are interchangeable.

If this were a rule of the system, it could speed up many derivations considerably. But if we *prove* that it is correct, isn't that good enough? Can't we use it just like a rule of inference, from then on?

There is no reason to doubt the correctness of this particular derived rule. But once you start admitting derived rules as part of your procedure in the Propositional Calculus, you have lost the formality of the system, since derived rules are derived informally-outside the system. Now formal systems were proposed as a way to exhibit every step of a proof explicitly, within one single, rigid framework, so that any mathematician could check another's work mechanically. But if you are willing to step outside of that framework at the drop of a hat, you might as well never have created it at all. Therefore, there is a drawback to using such shortcuts.

Formalizing Higher Levels

On the other hand, there is an alternative way out. Why not formalize the metatheory, too? That way, derived rules (metatheorems) would be theorems of a larger formal system, and it would be legitimate to look for shortcuts and derive them as theorems-that is, theorems of the formalized metatheory-which could then be used to speed up the derivations of theorems of the Propositional Calculus. This is an interesting idea, but as soon as it is suggested, one jumps ahead to think of metametatheories, and so on. It is clear that no matter how many levels you formalize, someone will eventually want to make shortcuts in the top level.

It might even be suggested that a theory of reasoning could be identical to its own metatheory, if it were worked out carefully. Then, it might seem, all levels would collapse into one, and thinking *about* the system would be just one way of working in the system! But it is not that easy. Even if a system can "think about itself", it still is not *outside* itself. You, outside the system, perceive it differently from the way it perceives itself. So there still is a metatheory-a view from outside-even for a theory which can "think about itself" inside itself. We will find that there are theories which can "think about themselves". In fact, we will soon see a system in which this happens completely accidentally, without our even intending it! And we will see what kinds of effects this produces. But for our study of the Propositional Calculus, we will stick with the simplest ideas-no mixing of levels.

Fallacies can result if you fail to distinguish carefully between working in the system (the **M**-mode) and thinking about the system (the **I**-mode). For example, it might seem perfectly reasonable to assume that, since $\mathbf{<P \vee \sim P>}$ (whose semi-interpretation is "either **P** or not **P**") is a theorem, either **P** or ~**P** must be a theorem. But this is dead wrong: neither one of the latter pair is a theorem. In general, it is a dangerous practice to assume that symbols can be slipped back and forth between different levels-here, the language of the formal system and its metalanguage (English).

Reflections on the Strengths and Weaknesses of the System

You have now seen one example of a system with a purpose-to re part of the architecture of logical thought. The concepts which this handles are very few in number, and they are very simple, precise co But the simplicity and precision of the Propositional Calculus are the kinds of features which make it appealing to mathematicians. There are two reasons for this. (1) It can be studied for its own properties, ex geometry studies simple, rigid shapes. Variants can be made on it, employing different symbols, rules of inference, axioms or axiom schemata on. (Incidentally, the version of the Propositional Calculus here pr is related to one invented by G. Gentzen in the early 1930's. The other versions in which only one rule of inference is used-detachment usually-and in which there are several axioms, or axiom schemata study of ways to carry out propositional reasoning in elegant formal systems is an appealing branch of pure mathematics. (2) The Propositional Calculus can easily be extended to include other fundamental aspects of reasoning. Some of this will be shown in the next Chapter, where the Propositional Calculus is incorporated lock, stock and barrel into a much larger and deeper system in which sophisticated number-theoretical reasoning can be done.

Proofs *vs.* Derivations

The Propositional Calculus is very much like reasoning in some w one should not equate its rules with the rules of human thought. A *proof* is something informal, or in other words a product of normal thought written in a human language, for human consumption. All sorts of complex features of thought may be used in proofs, and, though they may “feel right", one may wonder if they can be defended logically. That is really what formalization is for. A *derivation* is an artificial counterpart of and its purpose is to reach the same goal but via a logical structure whose methods are not only all explicit, but also very simple.

If -- and this is usually the case -it happens that a formal derivation is extremely lengthy compared with the corresponding "natural" proof that is just too bad. It is the price one pays for making each step so simple. What often happens is that a derivation and a proof are "simple" in complementary senses of the word. The proof is simple in that each step sounds right", even though one may not know just why; the derivation is simple in that each of its myriad steps is considered so trivial that it is beyond reproach, and since the whole derivation consists just of such trivial steps it is supposedly error-free. Each type of simplicity, however, brings along a characteristic type of complexity. In the case of proofs, it is the complexity of the underlying system on which they rest -- namely, human language -- and in the case of derivations, it is their astronomical size, which makes them almost impossible to grasp.

Thus, the Propositional Calculus should be thought of as part of a

general method for synthesizing artificial proof-like structures. It does not, however, have much flexibility or generality. It is intended only for use in connection with mathematical concepts-which are themselves quite rigid. As a rather interesting example of this, let us make a derivation in which a very peculiar string is taken as a premise in a fantasy: **<P∧~P>.** At least its semi-interpretation is peculiar. The Propositional Calculus, however, does not think about semi-interpretations; it just manipulates strings typographically-and typographically, there is really nothing peculiar about this string. Here is a fantasy with this string as its premise:

(1)	[	push
(2)	**<P∧~P>**	premise
(3)	**P**	separation
(4)	**~P**	separation
(5)	[	push
(6)	~Q	premise
(7)	**P**	carry-over line 3
(8)	**~~P**	double-tilde
(9)	]	pop
(10)	**<~Q⊃~~P>**	fantasy
(11)	**<~P⊃Q>**	contrapositive
(12)	**Q**	detachment (Lines 4,11)
(13)	]	pop
(14)	**<<P∧~P >⊃Q>**	fantasy

Now this theorem has a very strange semi-interpretation:

P and not **P** together imply **Q**

Since **Q** is interpretable by any statement, we can loosely take the theorem to say that "From a contradiction, anything follows"! Thus, in systems based on the Propositional Calculus, contradictions cannot be contained; they infect the whole system like an instantaneous global cancer.

The Handling of Contradictions

This does not sound much like human thought. If you found a contradiction in your own thoughts, it's very unlikely that your whole mentality would break down. Instead, you would probably begin to question the beliefs or modes of reasoning which you felt had led to the contradictory thoughts. In other words, to the extent you could, you would step out of the systems inside you which you felt were responsible for the contradiction, and try to repair them. One of the least likely things for you to do would be to throw up your arms and cry, "Well, I guess that shows that I believe everything now!" As a joke, yes-but not seriously.

Indeed, contradiction is a major source of clarification and progress in all domains of life-and mathematics is no exception. When in times past, a

contradiction in mathematics was found, mathematicians would immediately seek to pinpoint the system responsible for it, to jump out of it, to reason about it, and to amend it. Rather than weakening mathematics, the discovery and repair of a contradiction would strengthen it. This might take time and a number of false starts, but in the end it would yield fruit. For instance, in the Middle Ages, the value of the infinite series

1 – 1 + 1 – 1 + 1 -. ..

was hotly disputed. It was "proven" to equal 0, 1, ½, and perhaps other values. Out of such controversial findings came a fuller, deeper about infinite series.

A more relevant example is the contradiction right now confronting us-namely the discrepancy between the way we really think, and t the Propositional Calculus imitates us. This has been a source of discomfort for many logicians, and much creative effort has gone into trying to patch up the Propositional Calculus so that it would not act so stupidly and inflexibly. One attempt, put forth in the book *Entailment* by A. R. Anderson and N. Belnap,[3] involves "relevant implication", which tries to make the symbol for "if-then" reflect genuine causality, or at least connect meanings. Consider the following theorems of the Propositional Calculus

<P⊃<Q⊃P>>
<P⊃<Q∨~P>>
<<P∧~P>⊃Q>
<<P⊃Q>∨<Q⊃P>>

They, and many others like them, all show that there need be no relationship at all between the first and second clauses of an if-then statement for it to be provable within the Propositional Calculus. In protest, "relevant implication" puts certain restrictions on the contexts in which the rules of inference can be applied. Intuitively, it says that "something can only be derived from something else if they have to do with each other". For example, line 10 in the derivation given above would not be allowed in such a system, and that would block the derivation of the **<<P∧~P >⊃Q>**

More radical attempts abandon completely the quest for completeness or consistency, and try to mimic human reasoning with all its inconsistencies. Such research no longer has as its goal to provide a solid underpinning for mathematics, but purely to study human thought processes.

Despite its quirks, the Propositional Calculus has some feat recommend itself. If one embeds it into a larger system (as we will do next Chapter), and if one is sure that the larger system contains no contradictions (and we will be), then the Propositional Calculus does all that one could hope: it provides valid propositional inferences -- all that can be made. So if ever an incompleteness or an inconsistency is uncovered, can be sure that it will be the fault of the larger system, and not of its subsystem which is the Propositional Calculus.

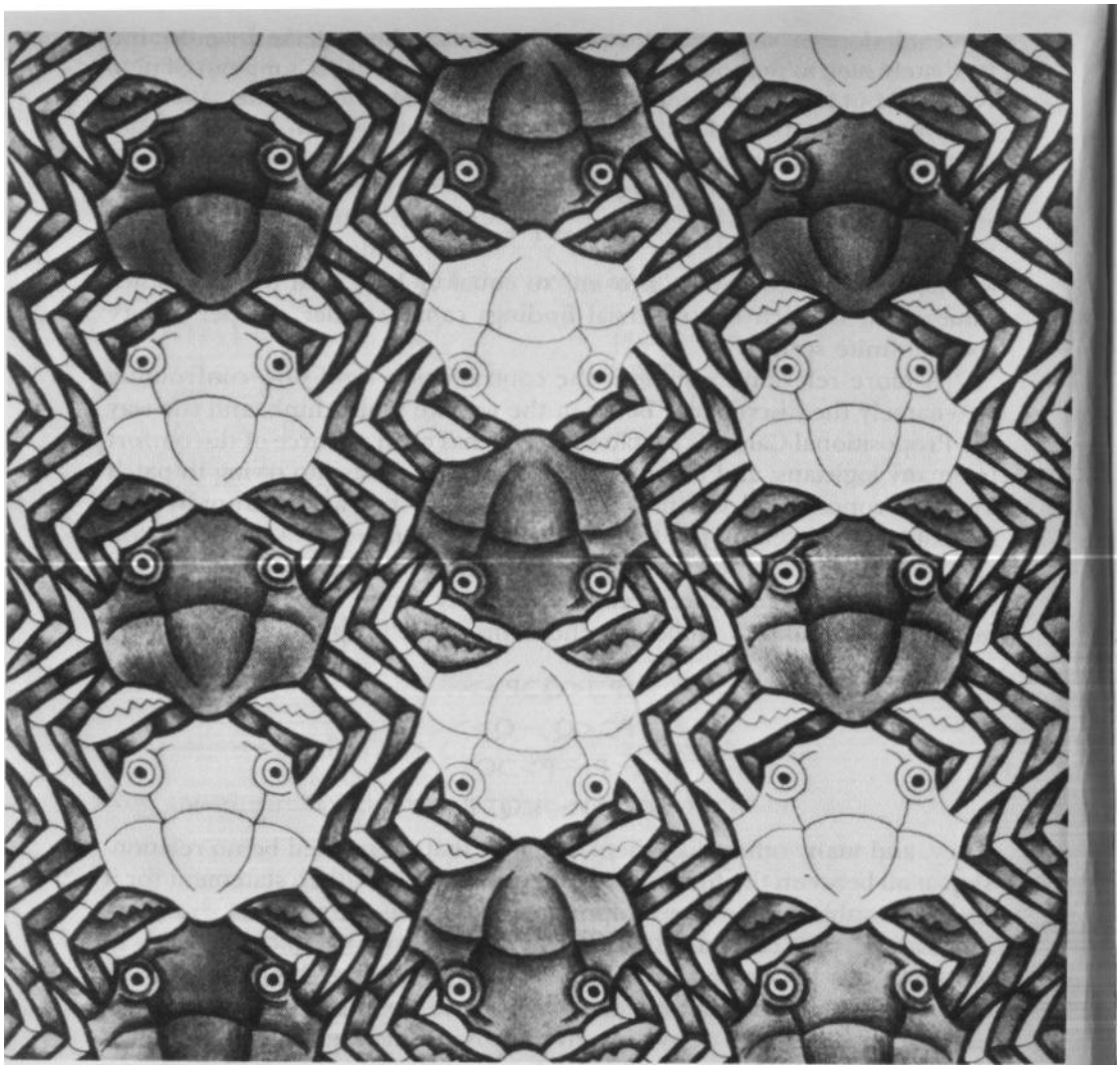

FIGURE 42. “Crab Canon”, by M. C. Escher (~1965)

Crab Canon

Achilles and the Tortoise happen upon each other in the park one day while strolling.

Tortoise: Good day, Mr. A.

Achilles: Why, same to you. Tortoise: So nice to run into you. Achilles: That echoes my thoughts.

Tortoise: And it's a perfect day for a walk. I think I'll be walking home soon.

Achilles: Oh, really? I guess there's nothing better for you than w Tortoise: Incidentally, you're looking in very fine fettle these days, I must say.

Achilles: Thank you very much.

Tortoise: Not at all. Here, care for one of my cigars?

Achilles: Oh, you are such a philistine. In this area, the Dutch contributions are of markedly inferior taste, don't you think?

Tortoise: I disagree, in this case. But speaking of taste, I finally saw that Crab Canon by your favorite artist, M. C. Escher, in a gallery the other day, and I fully appreciate the beauty and ingenuity with which he made one single theme mesh with itself going both backwards and forwards. But I am afraid I will always feel Bach is superior to Escher.

Achilles: I don't know. But one thing for certain is that I don't worry about arguments of taste. *De gustibus non est disputandum*.
Tortoise: Tell me, what's it like to be your age? Is it true that one has no worries at all?
Achilles: To be precise, one has no frets.
Tortoise: Oh, well, it's all the same to me.
Achilles: Fiddle. It makes a big difference, you know. Tortoise: Say, don't you play the guitar?
Achilles: That's my good friend. He often plays, the fool. But I myself wouldn't touch a guitar with a ten-foot pole!

(Suddenly, the Crab, appearing from out of nowhere, wanders up excitedly, pointing to a rather prominent black eye.)

Crab: Hallo! Hulloo! What's up? What's new? You see this bump, this lump? Given to me by a grump. Ho! And on such a fine day. You see, I was just idly loafing about the park when up lumbers this giant fellow from Warsaw-a colossal bear of a man-playing a lute. He was three meters tall, if I'm a day. I mosey on up to the chap, reach skyward and manage to tap him on the knee, saying, "Pardon me, sir, but you are Pole-luting our park with your mazurkas." But wow! he had no sense of humor-not a bit, not a wit-and POW!-he lets loose and belts me one, smack in the eye! Were it in my nature, I would crab up a storm, but in the time-honored tradition of my species, I backed off. After all, when we walk forwards, we move backwards. It's in our genes, you know, turning round and round. That reminds me-I've always wondered, "Which came first-the Crab, or the Gene?" That is to say, "Which came last the Gene, or the Crab?" I'm always turning things round and round, you know. It's in our genes, after all. When we walk backwards, we move forwards. Ah me, oh my! I must lope along on my merry way-so off I go on such a fine day. Sing "ho!" for the life of a Crab! TATA! iOle!

(And he disappears as suddenly as he arrived.)

Tortoise: That's my good friend. He often plays the fool. But I myself wouldn't touch a ten-foot Pole with a guitar!
Achilles: Say, don't you play the guitar? Tortoise: Fiddle. It makes a big difference, Achilles: Oh, well, it's all the same to me.
Tortoise: To be precise, one has no frets.
Achilles: Tell me, what's it like to be your age? Is it true that one has no worries at all?
Tortoise: I don't know. But one thing for certain is that I don't worry about arguments of taste. *Disputandum non est de gustibus.*

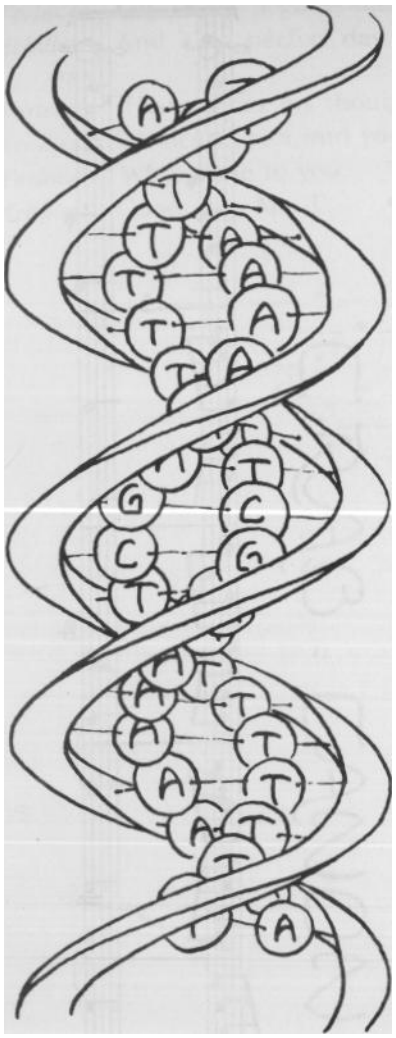

FIGURE 43. Here is a short section one of the Crab's Genes, turning round and round. When the two DNA strands are raveled and laid out side by side, they read this way:

….TTTTTTTTTCGAAAAAAAAA
….AAAAAAAAGCTTTTTTTTT

Notice that they are the same, only one forwards while the other goes backwards This is the defining property of the form called "crab canon" in music. It is reminiscent of, though a little different from palindrome, which is a sentence that reads the same backwards and forwards ,In molecular biology, such segments of ***DNA*** *are called "palindromes "-a slight misnomer, since "crab canon" would be more accurate. Not only is this* ***DNA*** *segment crab-canonical-but moreover its base sequence codes for the Dialogue's structure Look carefully!*

...

Achilles: I disagree, in this case. But speaking of taste, I finally heard that *Crab Canon* by your favorite composer, J. S. Bach, in a concert other day, and I fully appreciate the beauty and ingenuity with which he made one single theme mesh with itself going both backwards and forwards. But I'm afraid I will always feel Escher is superior to Bach

Tortoise: Oh, you are such a philistine. In this area, the Dutch contributions are of markedly inferior taste, don't you think?

Achilles: Not at all. Here, care for one of my cigars?

Tortoise: Thank you very much.

Achilles: Incidentally, you're looking in very fine fettle these days, I must say.

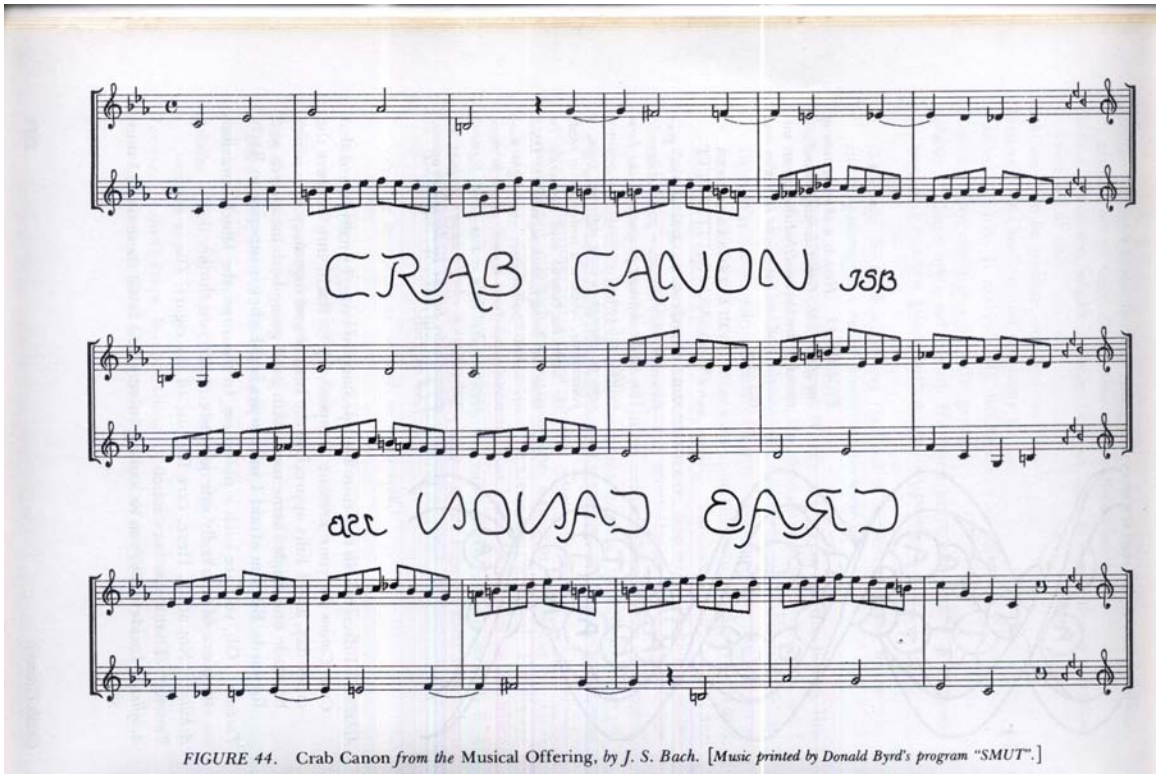

FIGURE 44. Crab Canon *from the* Musical Offering, *by J. S. Bach.* [*Music printed by Donald Byrd's program "SMUT".*]

Tortoise: Oh, really? I guess there's nothing better for you than walking.
Achilles: And it's a perfect day for a walk. I think I'll be walking home soon.
Tortoise: That echoes my thoughts.
Achilles: So nice to run into you.
Tortoise: Why, same to you.
Achilles: Good day, Mr. T

CHAPTER VIII

Typographical Number Theory

The *Crab Canon* and Indirect Self-Reference

THREE EXAMPLES OF indirect self-reference are found in the Crab Canon. Achilles and the Tortoise both describe artistic creations they know-and, quite accidentally, those creations happen to have the same structure as the Dialogue they're in. (Imagine my surprise, when I, the author, noticed this!) Also, the Crab describes a biological structure and that, too, has the same property. Of course, one could read the Dialogue and understand it and somehow fail to notice that it, too, has the form of a crab canon. This would be understanding it on one level, but not on another. To see the self-reference, one has to look at the form, as well as the content, of the Dialogue.

Gödel's construction depends on describing the form, as well as the content, of strings of the formal system we shall define in this Chapter -- *Typographical Number* Theory (**TNT**). The unexpected twist is that, because of the subtle mapping which Gödel discovered, the form of strings can be described in the formal system itself. Let us acquaint ourselves with this strange system with the capacity for wrapping around.

What We Want to Be Able to Express in TNT

We'll begin by citing some typical sentences belonging to number theory; then we will try to find a set of basic notions in terms of which all our sentences can be rephrased. Those notions will then be given individual symbols. Incidentally, it should be stated at the outset that the term "number theory" will refer only to properties of positive integers and zero (and sets of such integers). These numbers are called the *natural numbers*. Negative numbers play no role in this theory. Thus the word "number", when used, will mean exclusively a natural number. And it is important -- vital-for you to keep separate in your mind the formal system (**TNT**) and the rather ill-defined but comfortable old branch of mathematics that is number theory itself; this I shall call "**N**".
Some typical sentences of **N**-number theory-are:

(1) 5 is prime.
(2) 2 is not a square.
(3) 1729 is a sum of two cubes.
(4) No sum of two positive cubes is itself a cube.
(5) There are infinitely many prime numbers.
(6) 6 is even.